

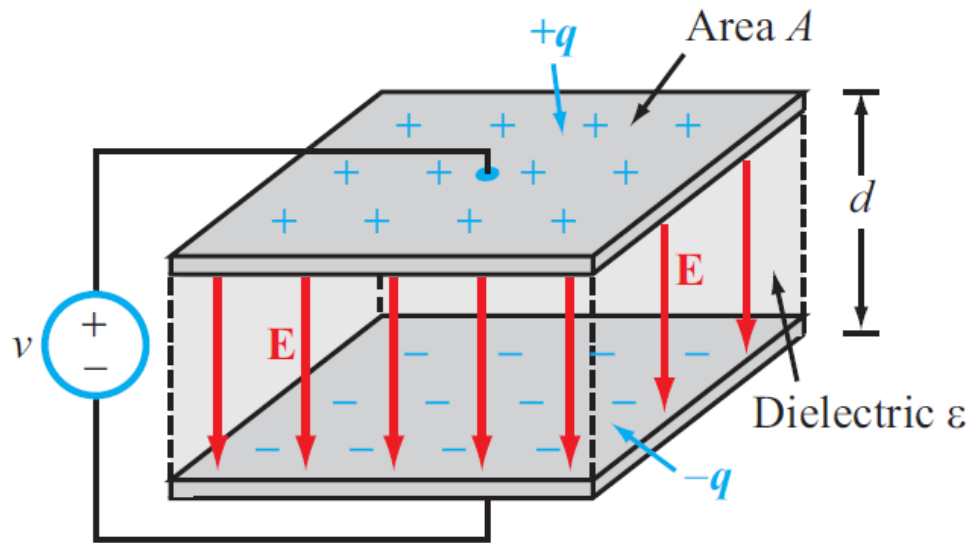


Lecture 5

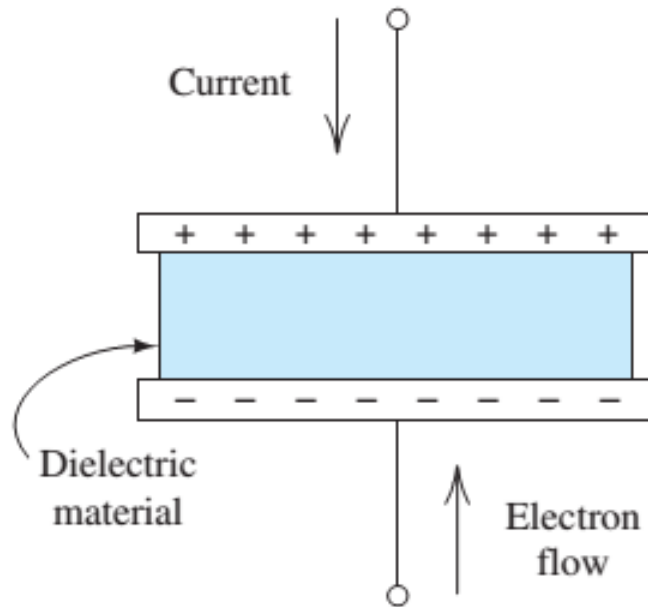
- Capacitors and Inductors

Capacitors

Storage element that stores energy in electric field



Parallel plate capacitor



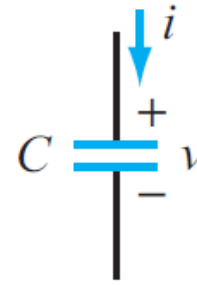
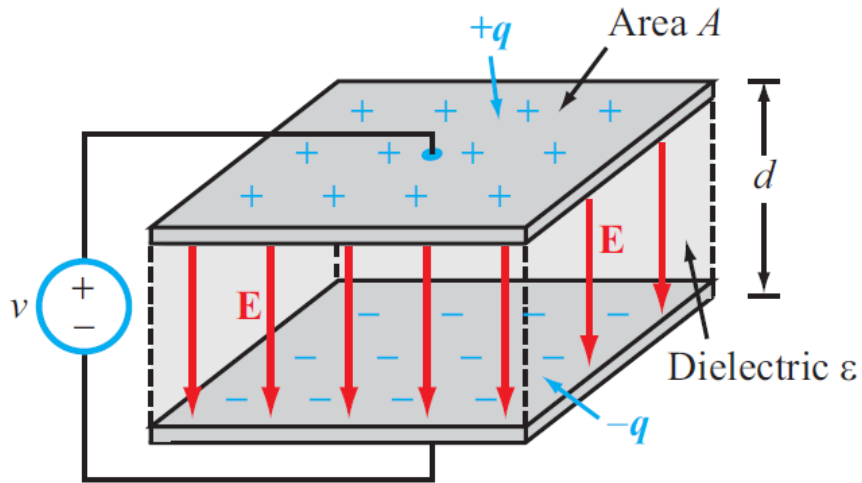
(a) As current flows through a capacitor, charges of opposite signs collect on the respective plates

Does DC voltage generate current flow through a capacitor?

Does AC voltage generate current flow through a capacitor?



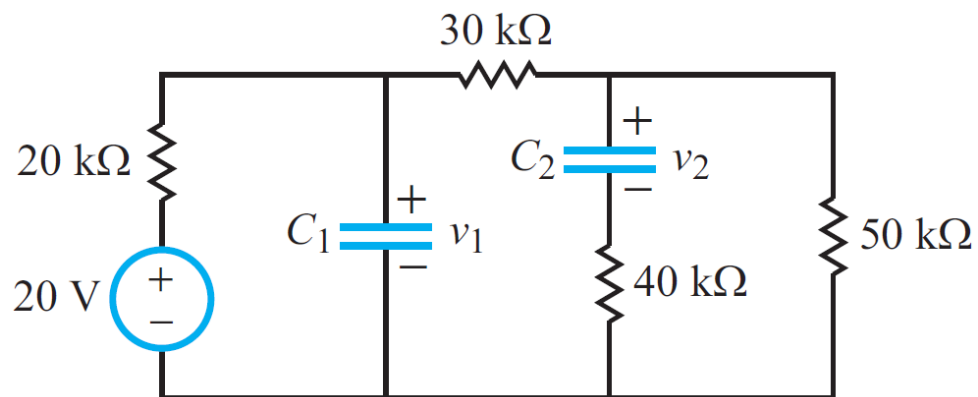
V-I Relationship of Capacitors



$$i = C \frac{dv}{dt}$$

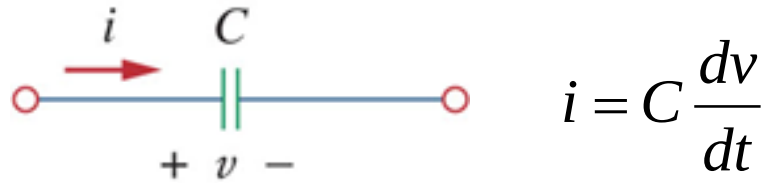


Example





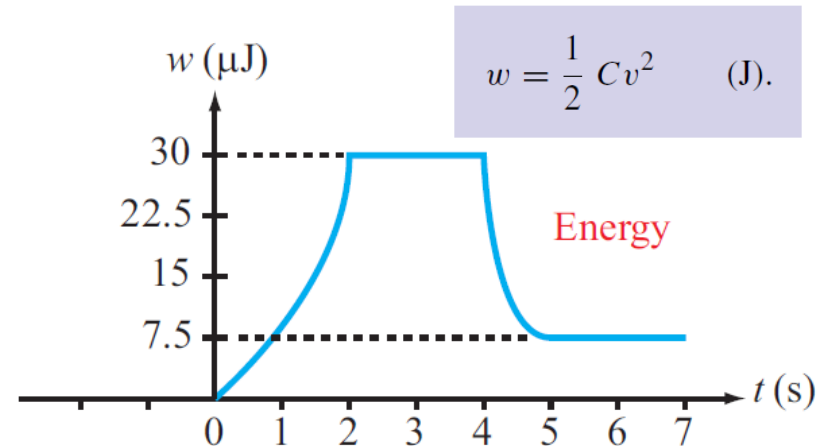
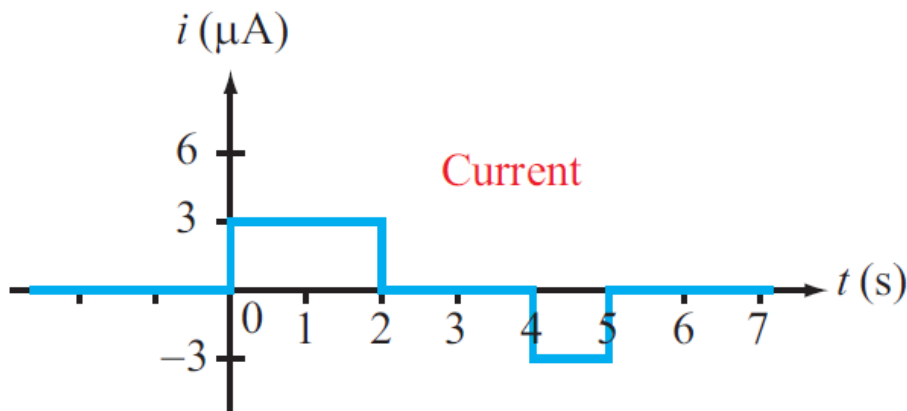
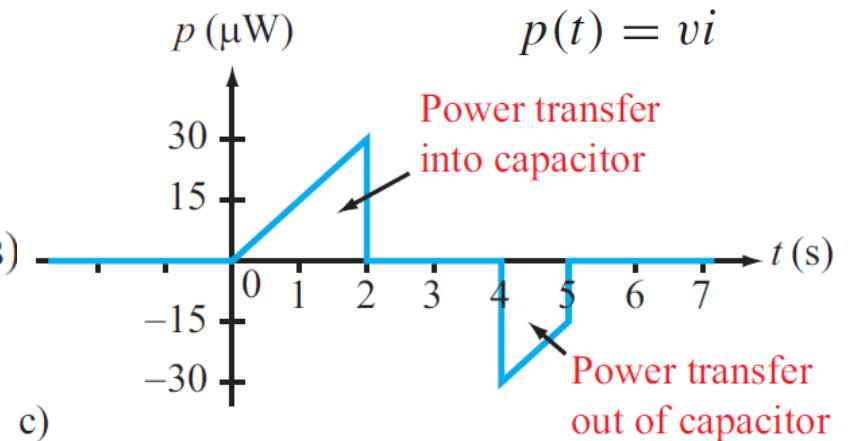
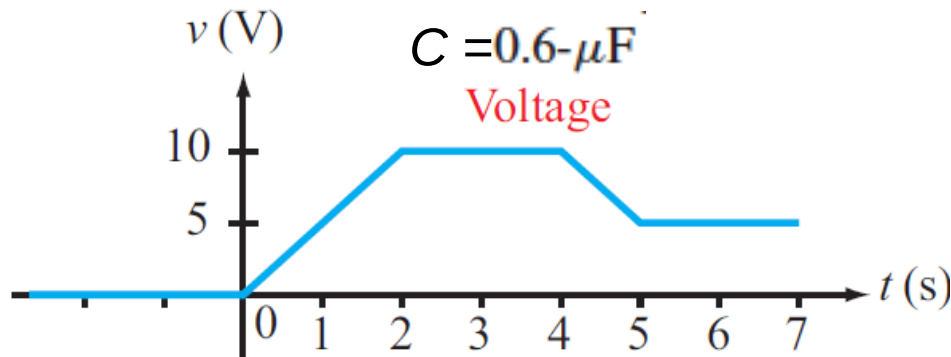
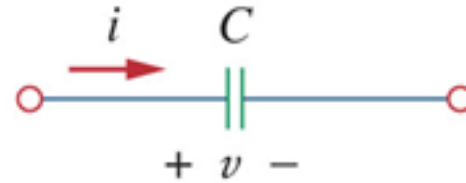
Stored Energy



- The instantaneous power delivered to the capacitor is
- The energy stored in a capacitor is:



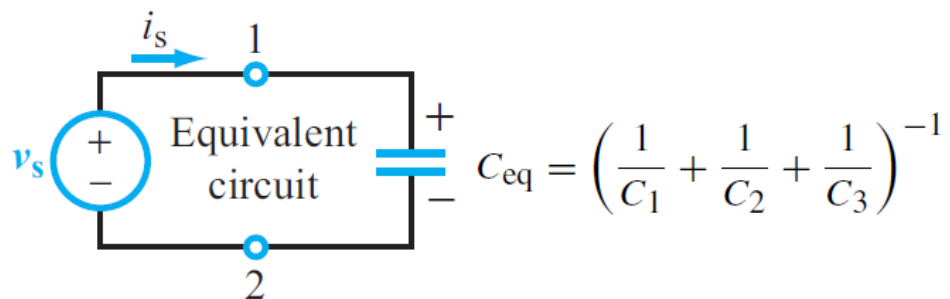
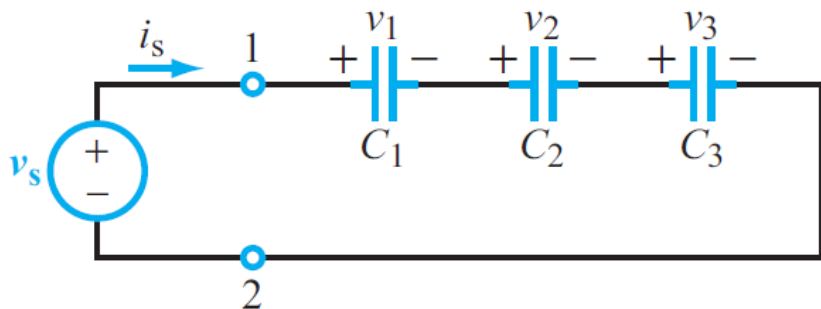
Capacitor Response





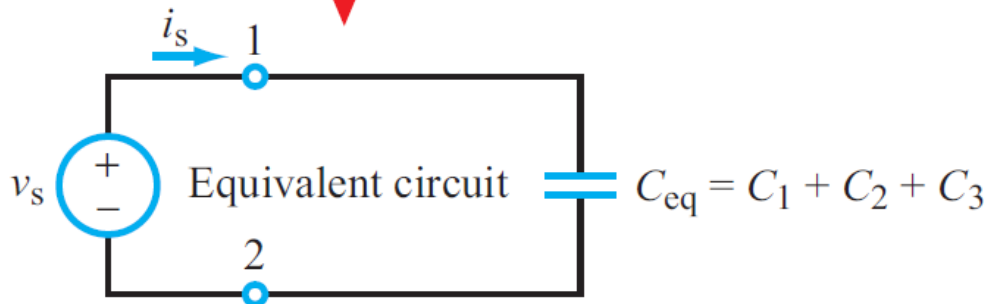
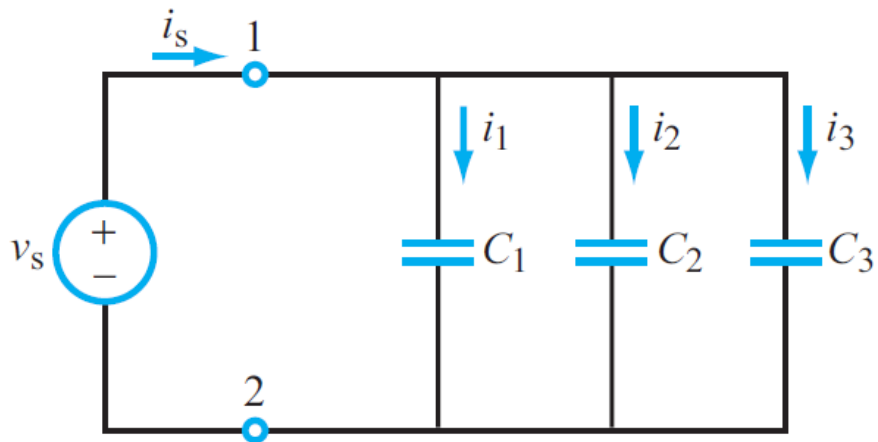
Capacitors in Series

Combining In-Series Capacitors



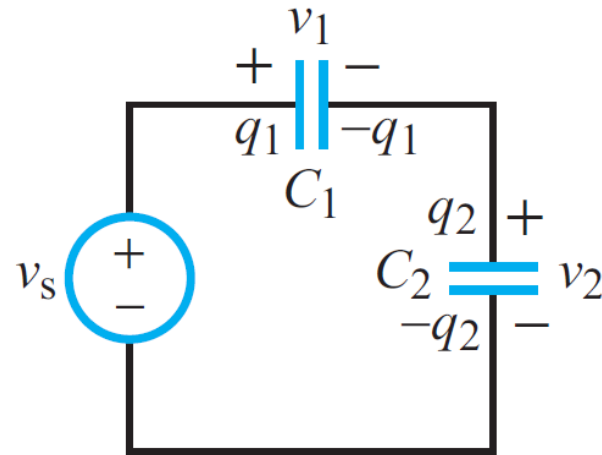


Capacitors in Parallel





Voltage Division

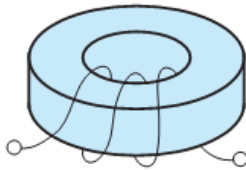


$$v_1 = \left(\frac{C_2}{C_1 + C_2} \right) v_s$$

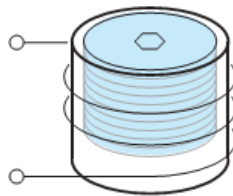
$$v_2 = \left(\frac{C_1}{C_1 + C_2} \right) v_s$$

Inductors

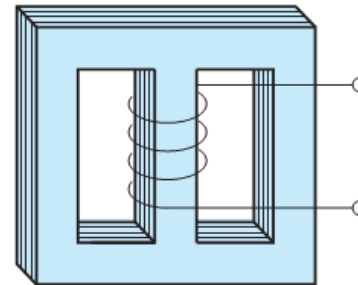
- A storage element that stores energy in magnetic field.
 - They have applications in power supplies, transformers, radios, TVs, radars, and electric motors.
- Any conductor has inductance, but the effect is typically enhanced by coiling the wire up.



(a) Toroidal inductor



(b) Coil with an iron-oxide slug that can be screwed in or out to adjust the inductance

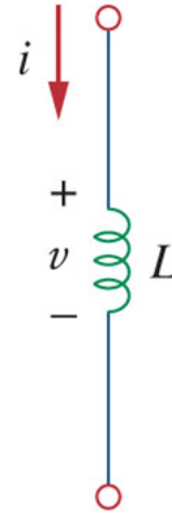
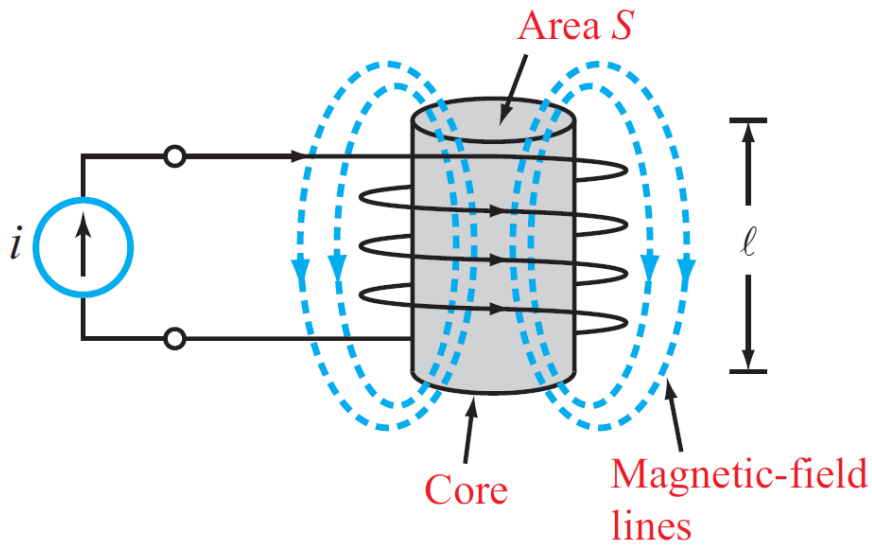


(c) Inductor with a laminated iron core

$$L = \frac{N^2 \mu S}{l}$$



V-I Relationship of Inductors

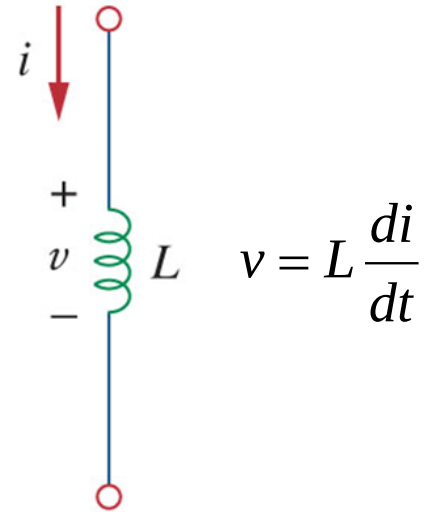


$$v = L \frac{di}{dt}$$



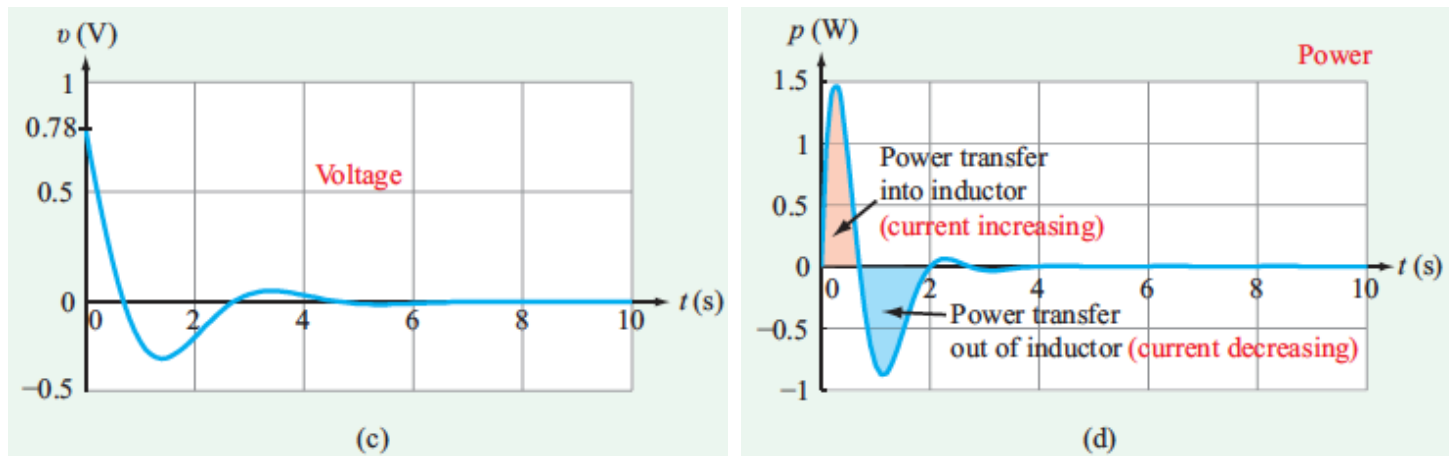
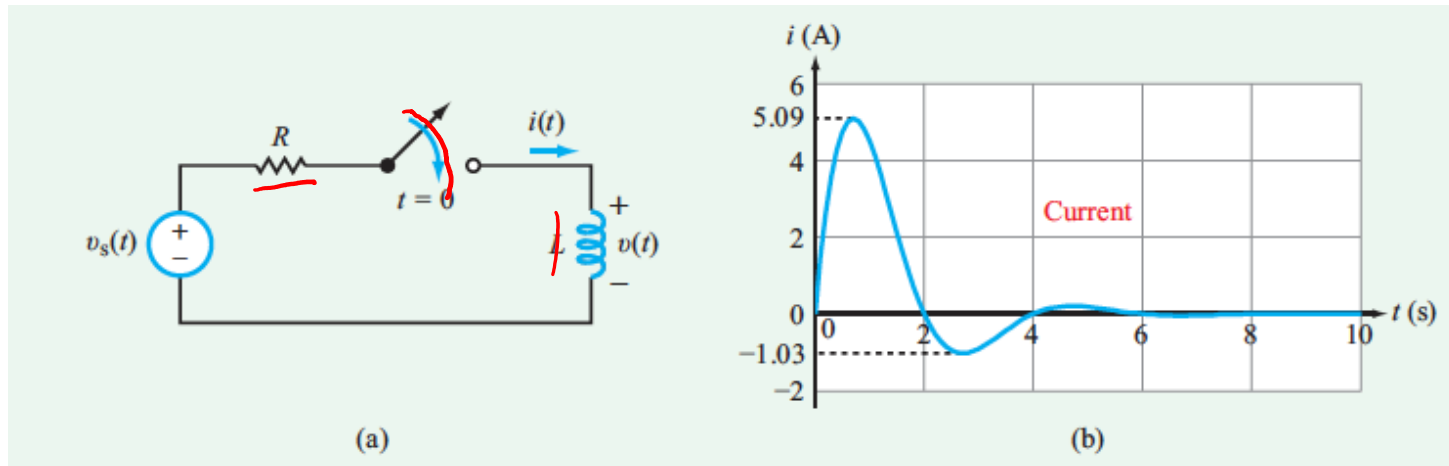
Energy Stored in an Inductor

- The power delivered to the inductor is:
- The energy stored is:



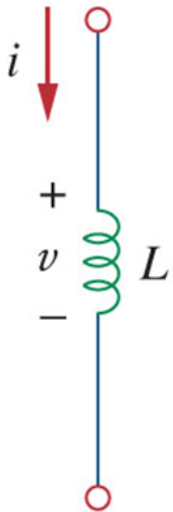
Inductor Response

$$i(t) = 10e^{-0.8t} \sin(\pi t/2) \text{ A}, \quad (\text{for } t \geq 0)$$



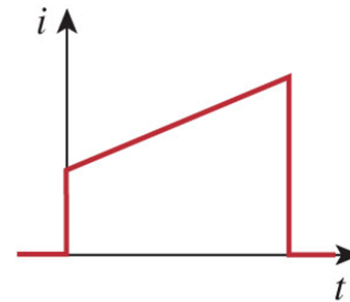
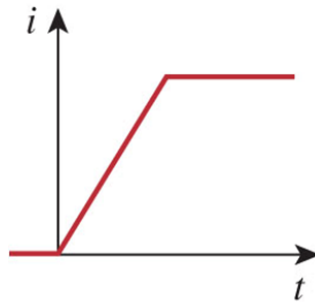


Important Property of Inductors



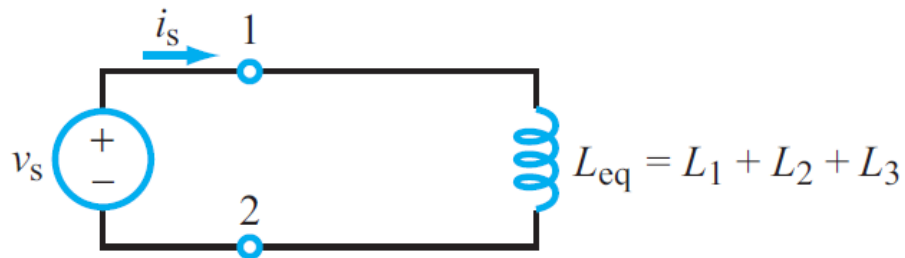
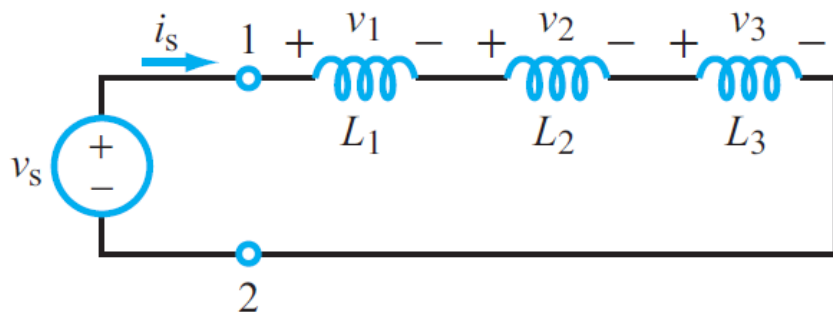
$$v = L \frac{di}{dt}$$

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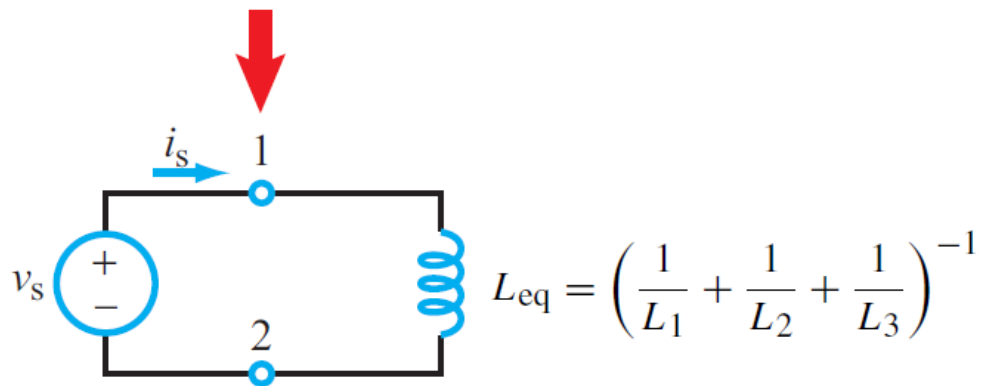
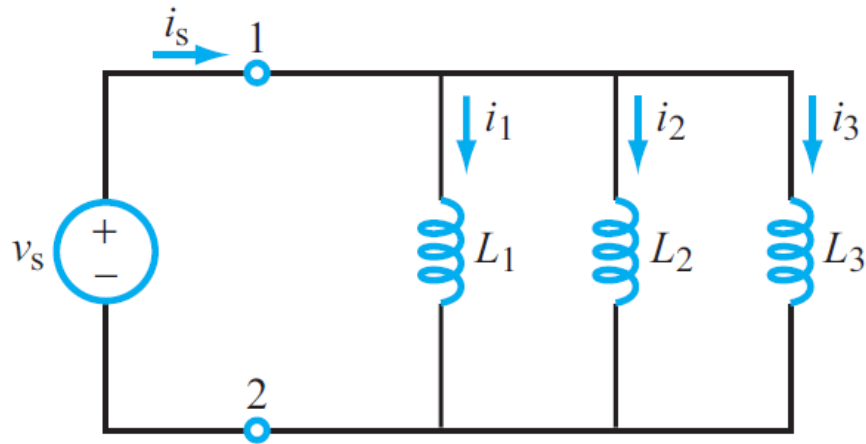
Inductors in Series





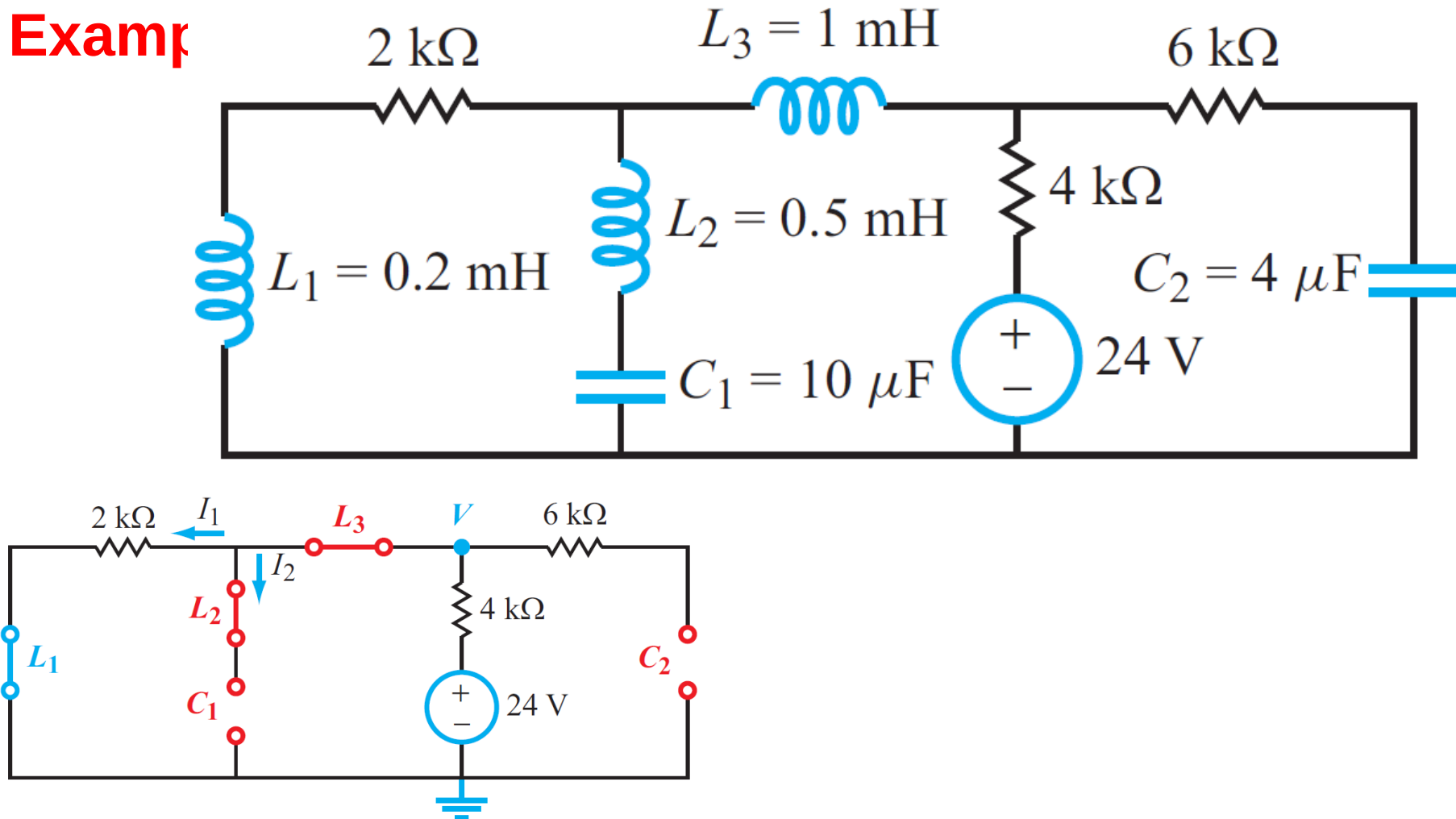
Inductors in Parallel

Combining In-Parallel Inductors



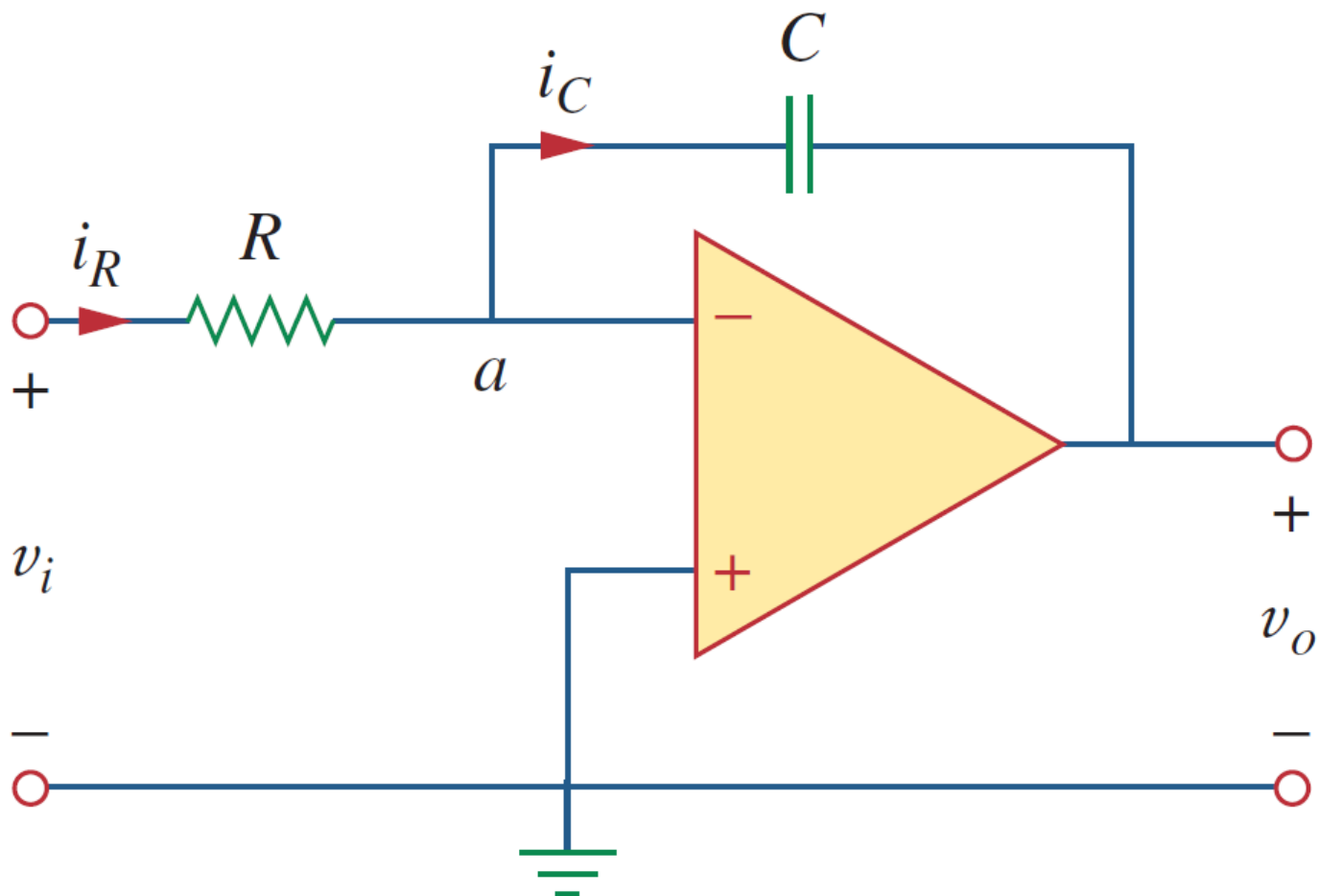


Example



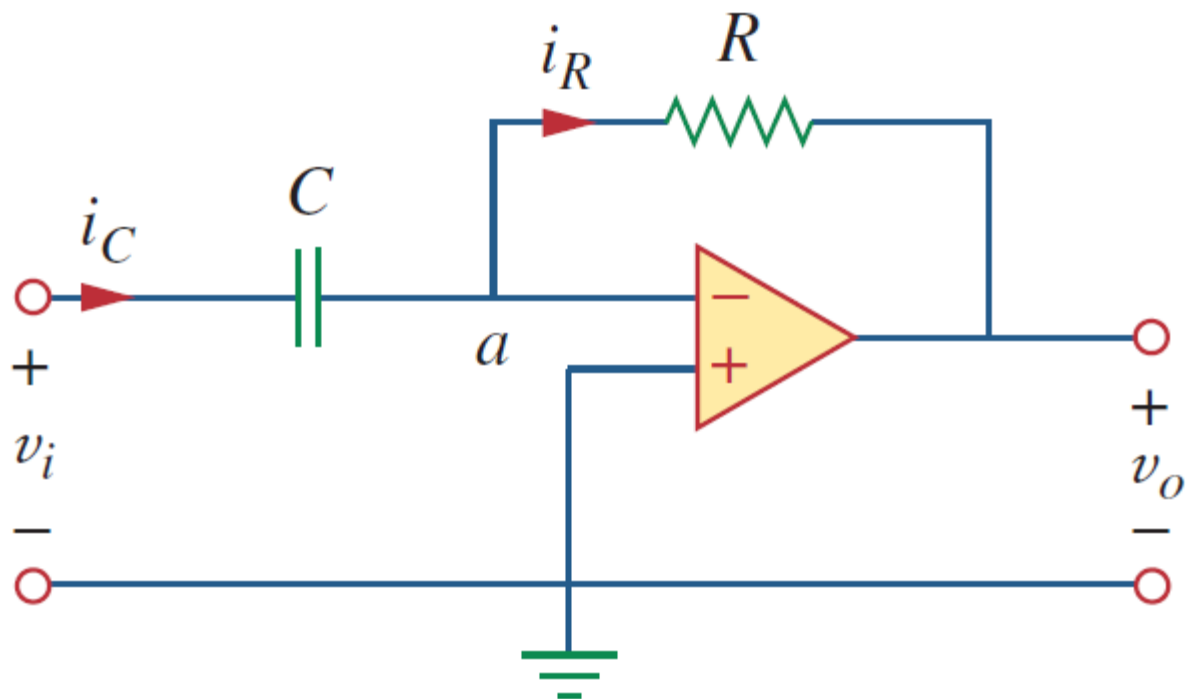


Integrator





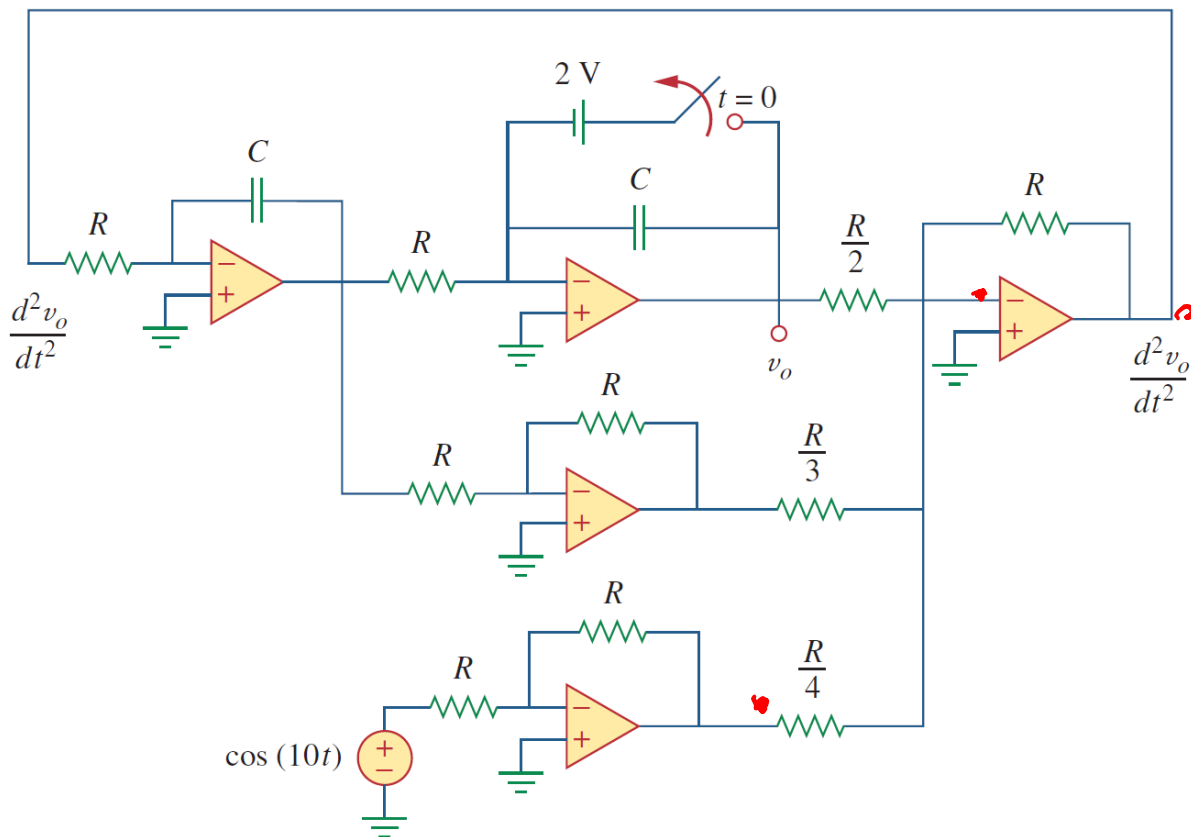
Differentiator



Analog Computer

$$\frac{d^2 v_o}{dt^2} + 3 \frac{dv_o}{dt} + 2v_o = 4 \cos 10t, \quad t > 0$$

subject to $v_o(0) = 2, v'_o(0) = 0$.





Summary of Capacitors and Inductors

Table 5-4: Basic properties of R , L , and C .

Property	R	L	C
i - v relation	$i = \frac{v}{R}$	$i = \frac{1}{L} \int_{t_0}^t v dt' + i(t_0)$	$i = C \frac{dv}{dt}$
v - i relation	$v = iR$	$v = L \frac{di}{dt}$	$v = \frac{1}{C} \int_{t_0}^t i dt' + v(t_0)$
p (power transfer in)	$p = i^2 R$	$p = Li \frac{di}{dt}$	$p = Cv \frac{dv}{dt}$
w (stored energy)	0	$w = \frac{1}{2} Li^2$	$w = \frac{1}{2} Cv^2$
Series combination	$R_{eq} = R_1 + R_2$	$L_{eq} = L_1 + L_2$	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$
Parallel combination	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$	$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$	$C_{eq} = C_1 + C_2$
dc behavior	no change	short circuit	open circuit
Can v change instantaneously?	yes	yes	no
Can i change instantaneously?	yes	no	yes